Delta Lake Incremental Processing Assignment

Objective: Perform incremental data processing using Delta Lake. The screenshots show dataset loading, cleaning, incremental data creation, MERGE operation, result validation, final output, and SCD Type 2 implementation.

Figure 1: Setup Code

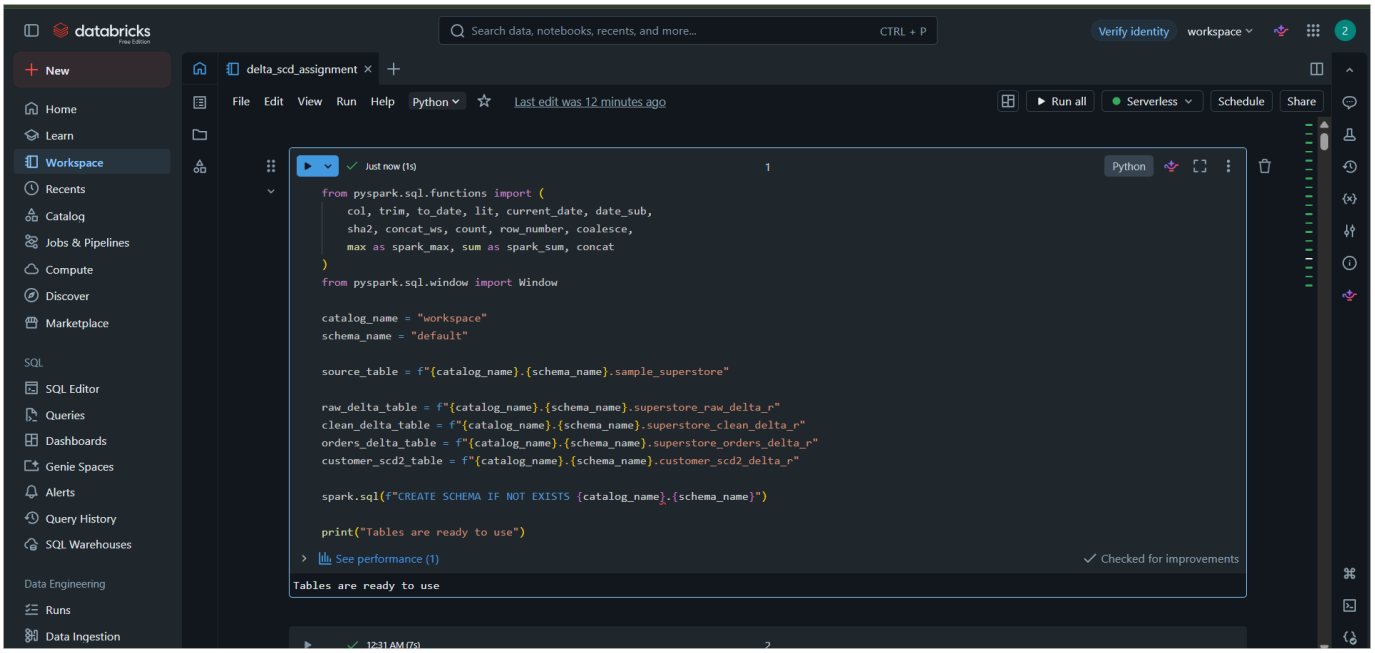
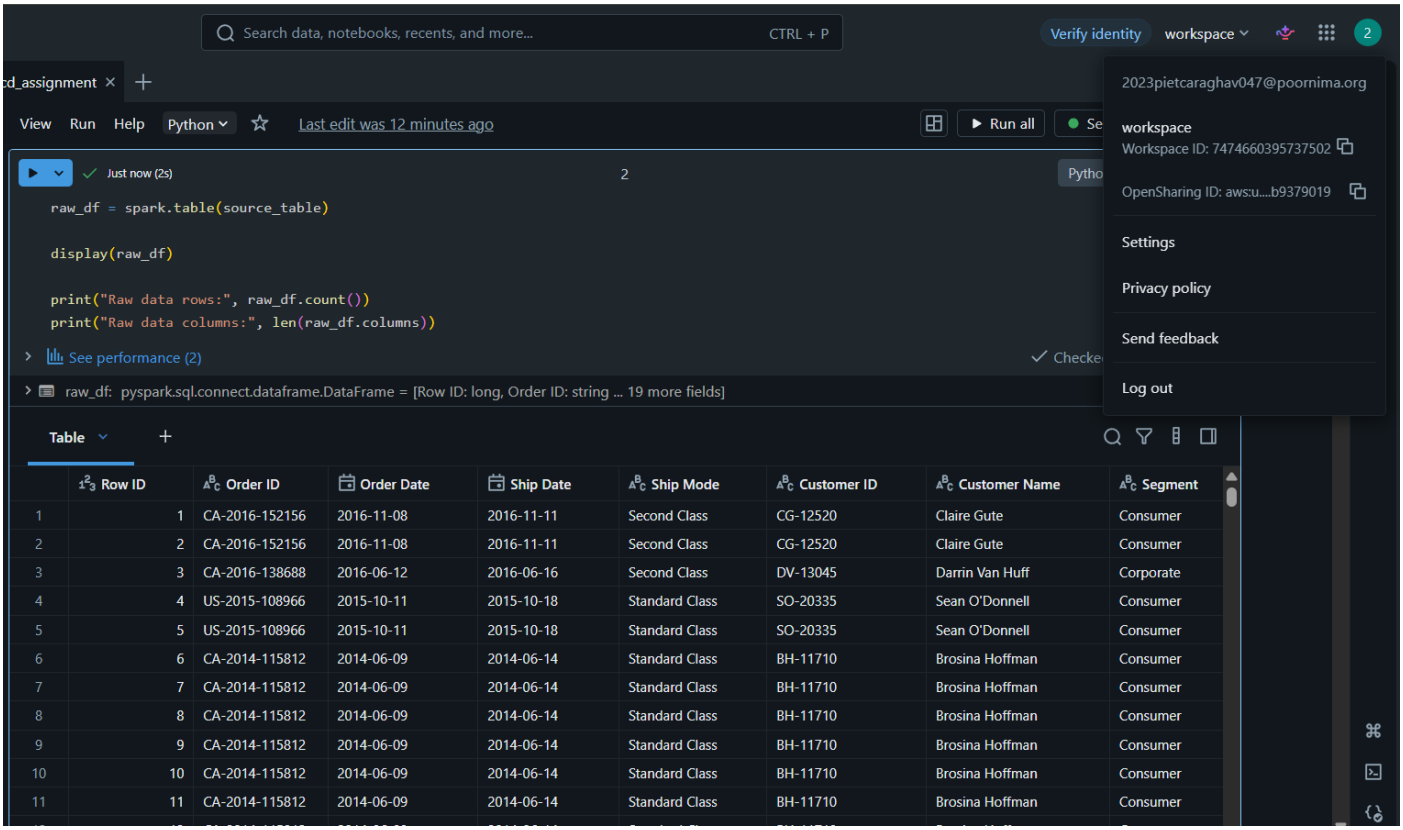


Figure 2: Raw Data



	Row ID	Order ID	Order Date	Ship Date	Ship Mode	Customer ID	Customer Name	Segment
1	1	CA-2016-152156	2016-11-08	2016-11-11	Second Class	CG-12520	Claire Gute	Consumer
2	2	CA-2016-152156	2016-11-08	2016-11-11	Second Class	CG-12520	Claire Gute	Consumer
3	3	CA-2016-138688	2016-06-12	2016-06-16	Second Class	DV-13045	Darrin Van Huff	Corporate
4	4	US-2015-108966	2015-10-11	2015-10-18	Standard Class	SO-20335	Sean O'Donnell	Consumer
5	5	US-2015-108966	2015-10-11	2015-10-18	Standard Class	SO-20335	Sean O'Donnell	Consumer
6	6	CA-2014-115812	2014-06-09	2014-06-14	Standard Class	BH-11710	Brosina Hoffman	Consumer
7	7	CA-2014-115812	2014-06-09	2014-06-14	Standard Class	BH-11710	Brosina Hoffman	Consumer
8	8	CA-2014-115812	2014-06-09	2014-06-14	Standard Class	BH-11710	Brosina Hoffman	Consumer
9	9	CA-2014-115812	2014-06-09	2014-06-14	Standard Class	BH-11710	Brosina Hoffman	Consumer
10	10	CA-2014-115812	2014-06-09	2014-06-14	Standard Class	BH-11710	Brosina Hoffman	Consumer
11	11	CA-2014-115812	2014-06-09	2014-06-14	Standard Class	BH-11710	Brosina Hoffman	Consumer

Figure 3: Data Check

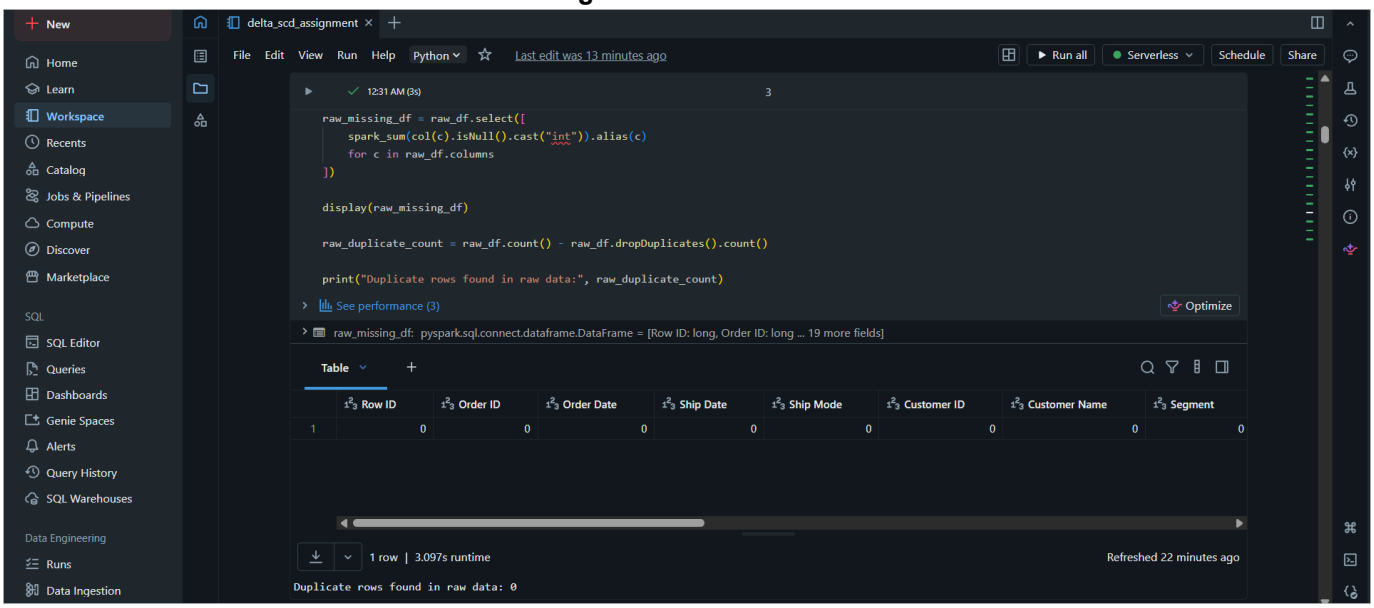


Figure 4: Column Cleanup

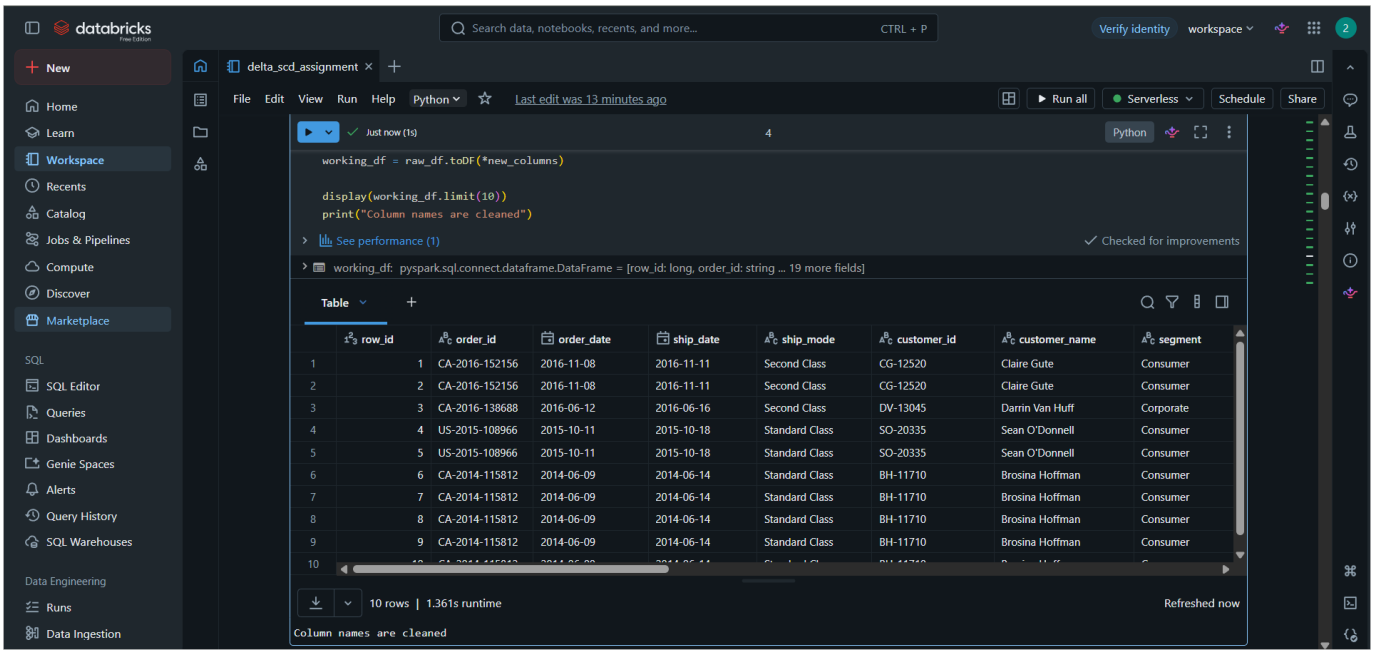


Figure 5: Incremental Logic

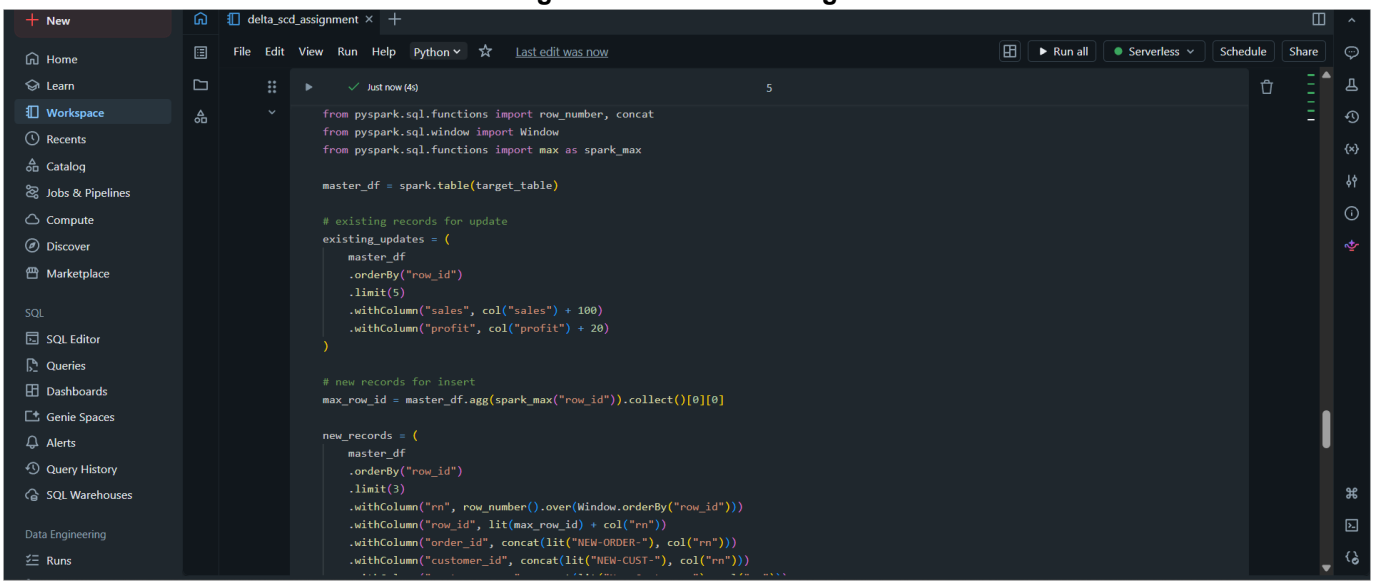


Figure 6: Incremental Output

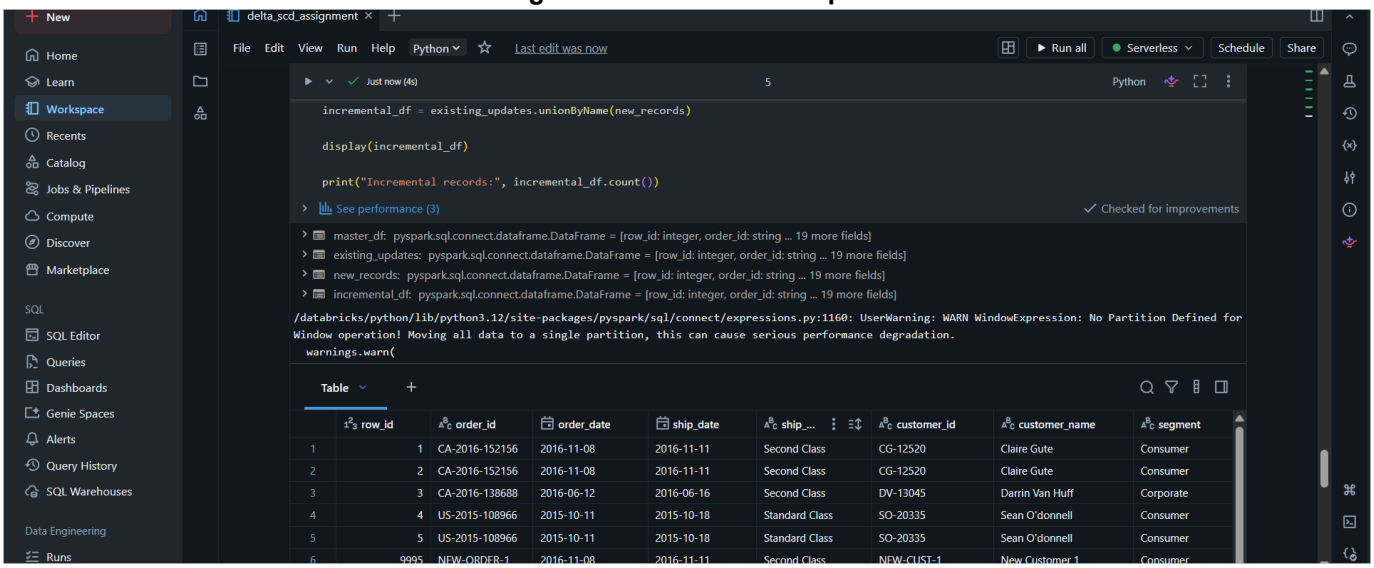


Figure 7: Merge Query

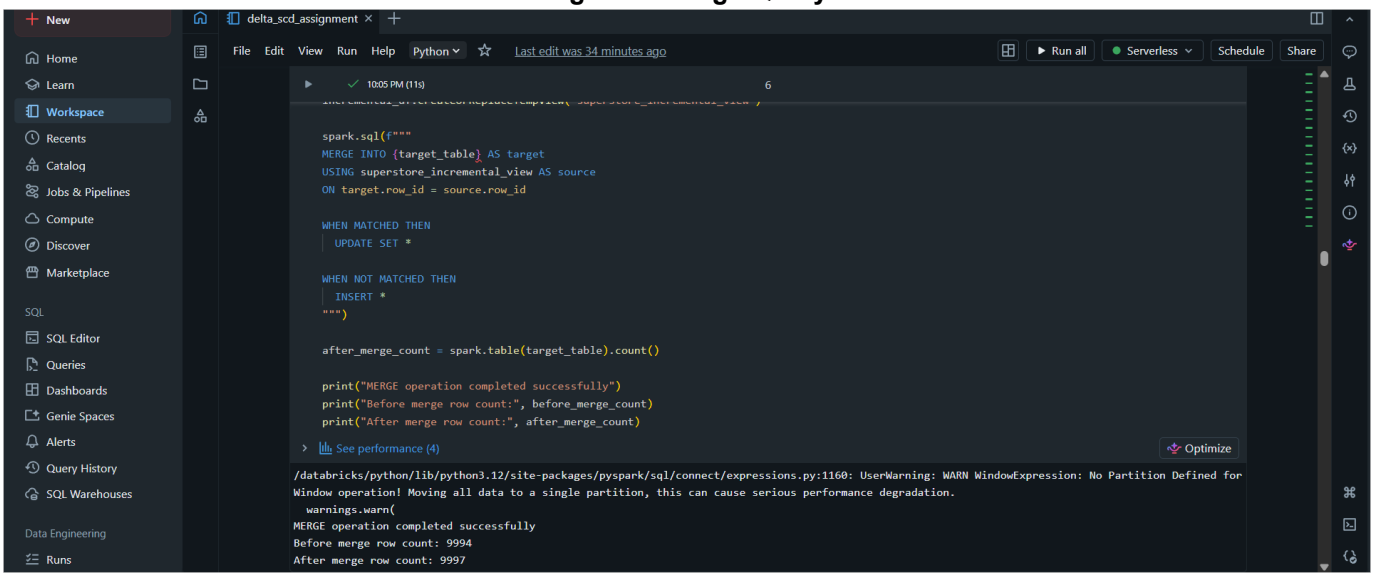


Figure 8: Validation Logic

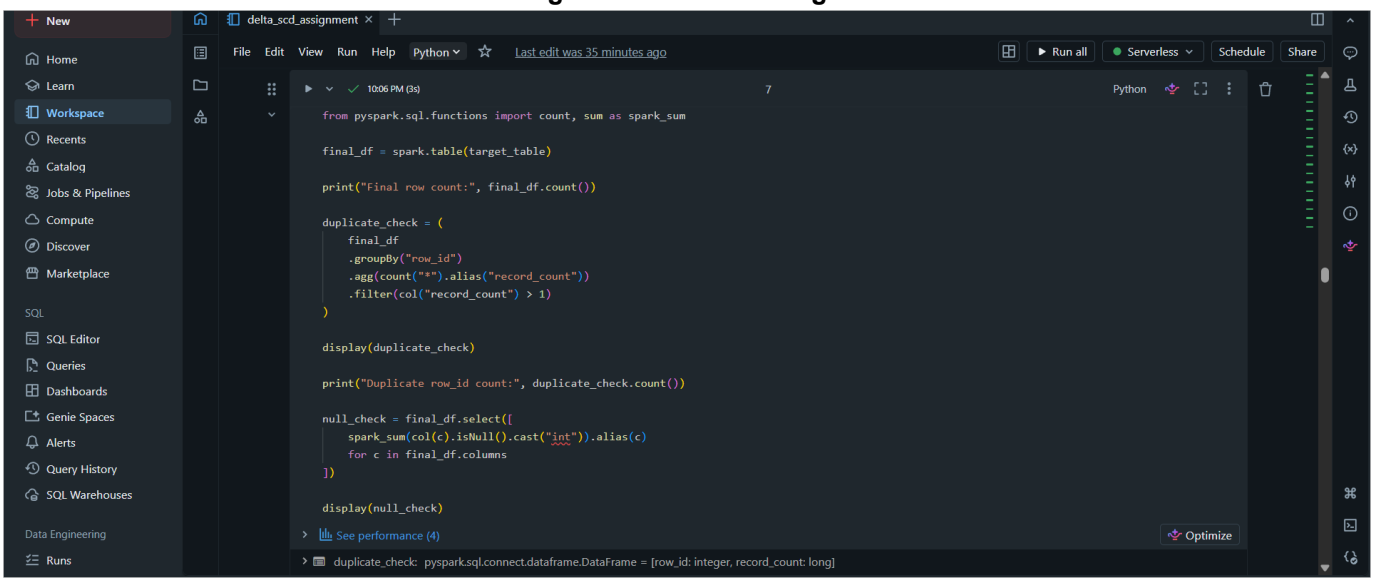


Figure 9: Duplicate Check

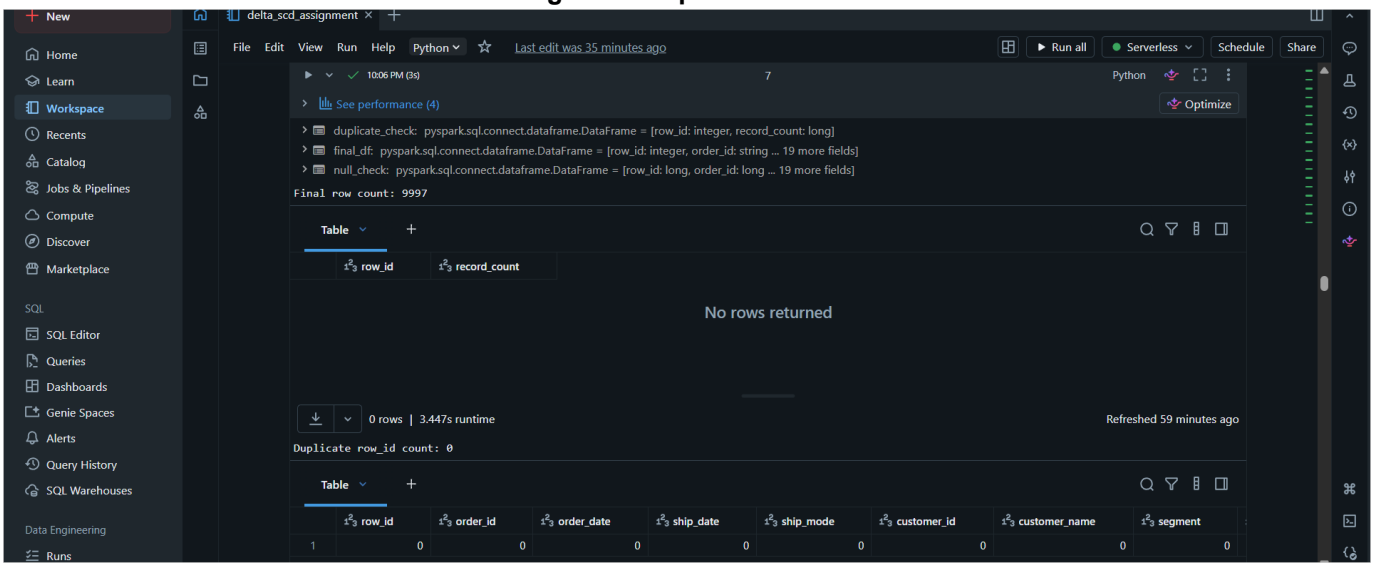


Figure 10: Final Data

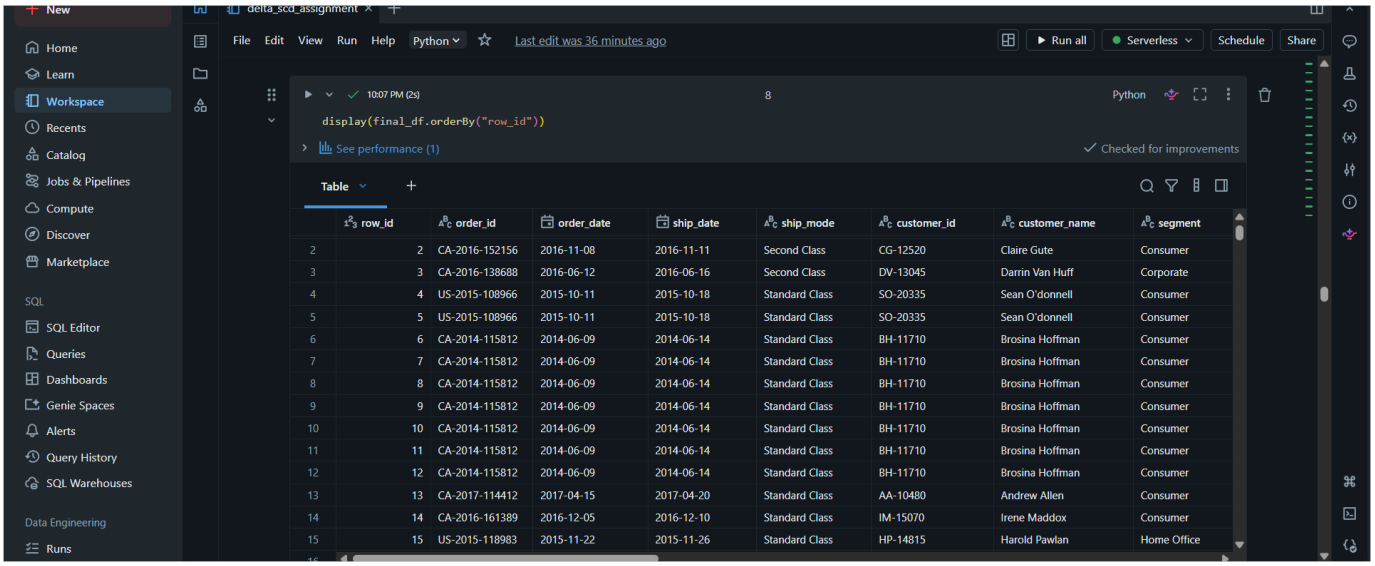


Figure 11: Summary Logic

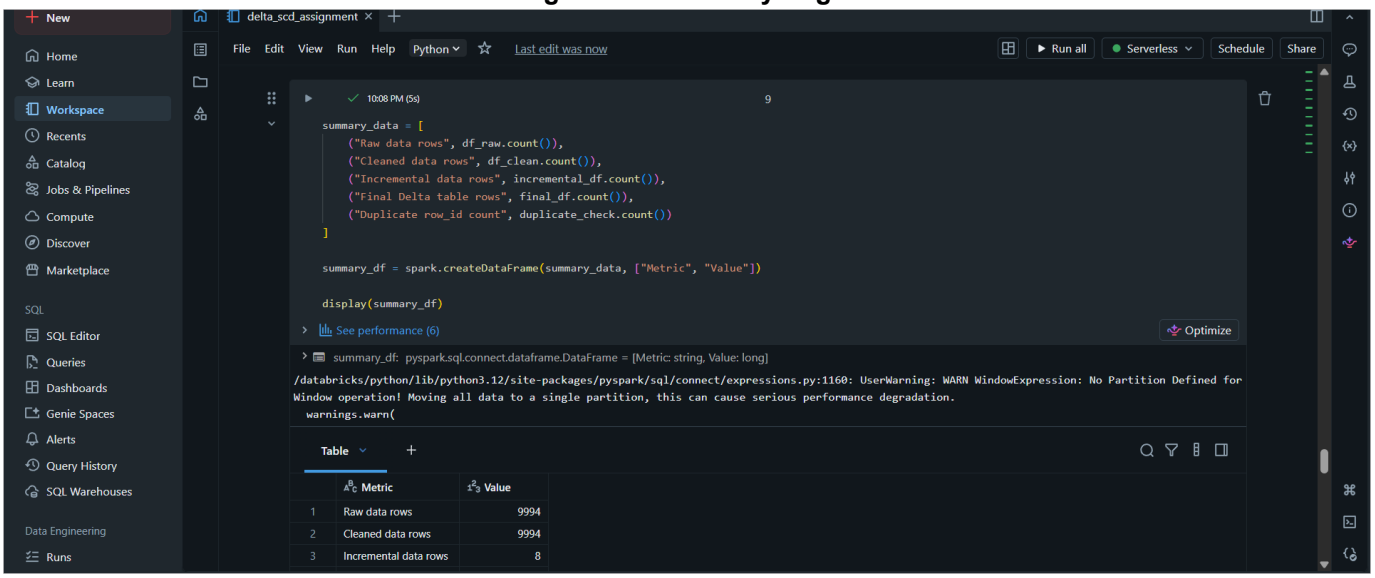


Figure 12: Summary Output

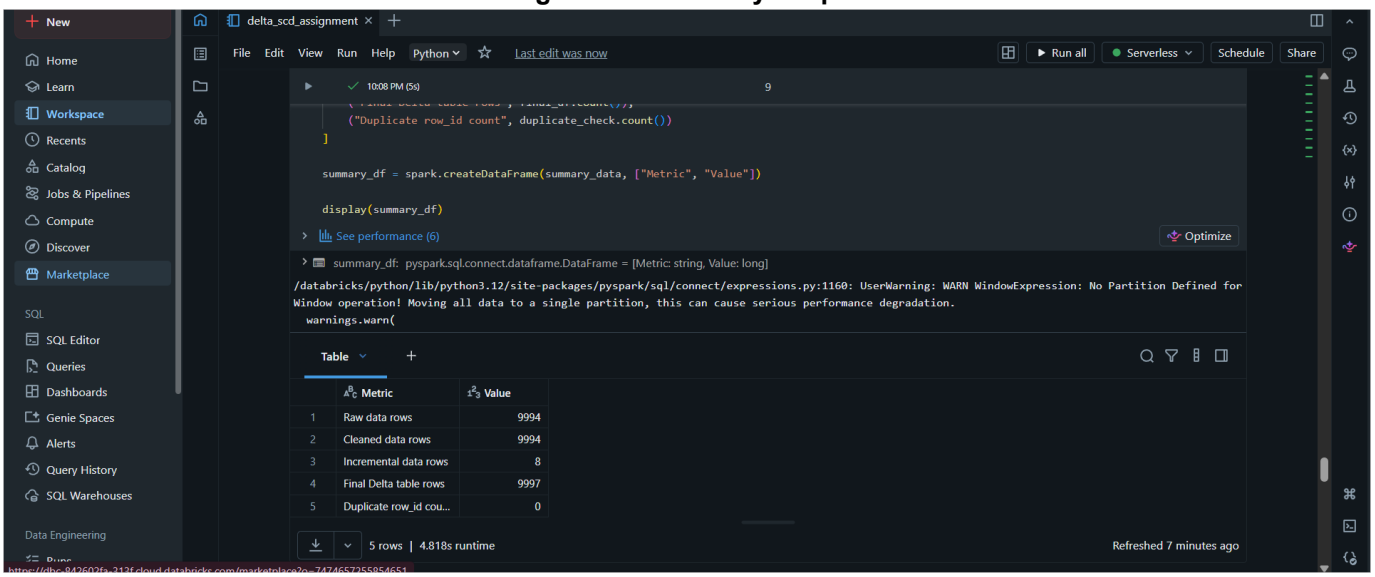


Figure 13: SCD2 Setup

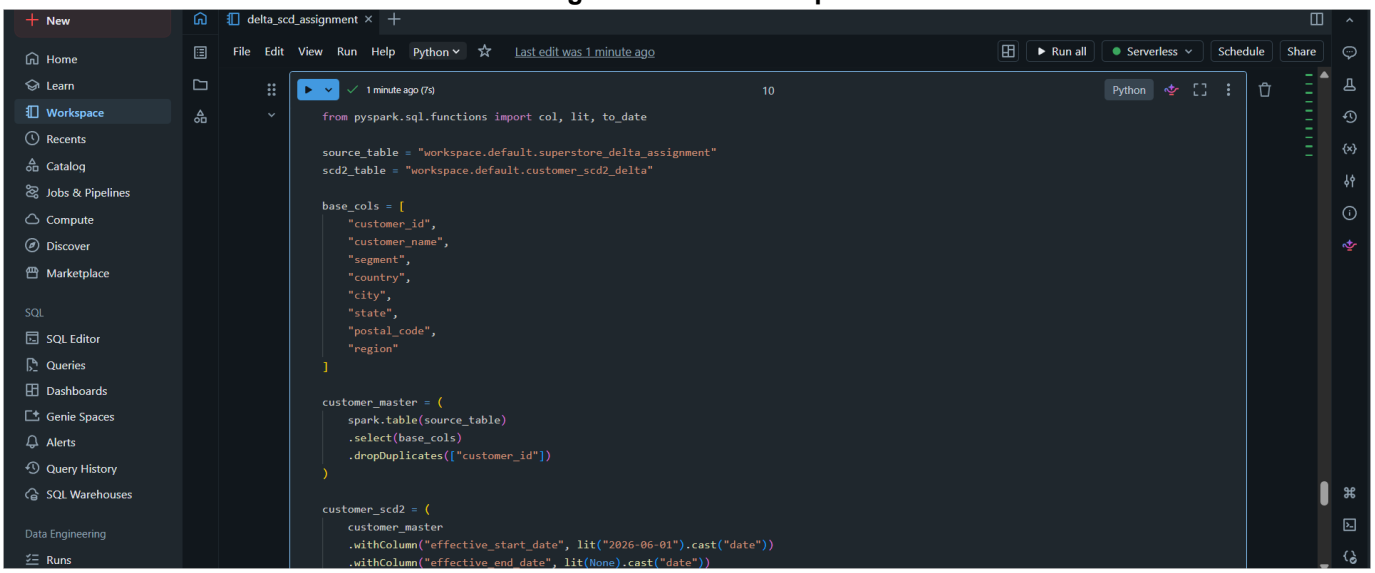


Figure 14: Initial SCD2

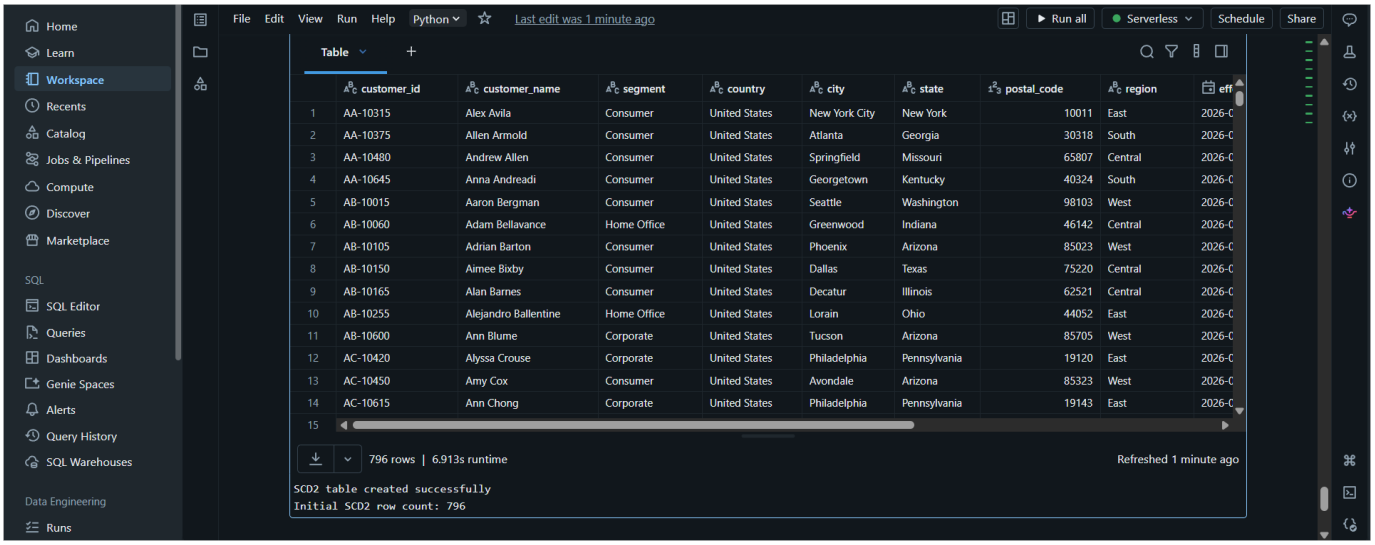


Figure 15: Customer Changes

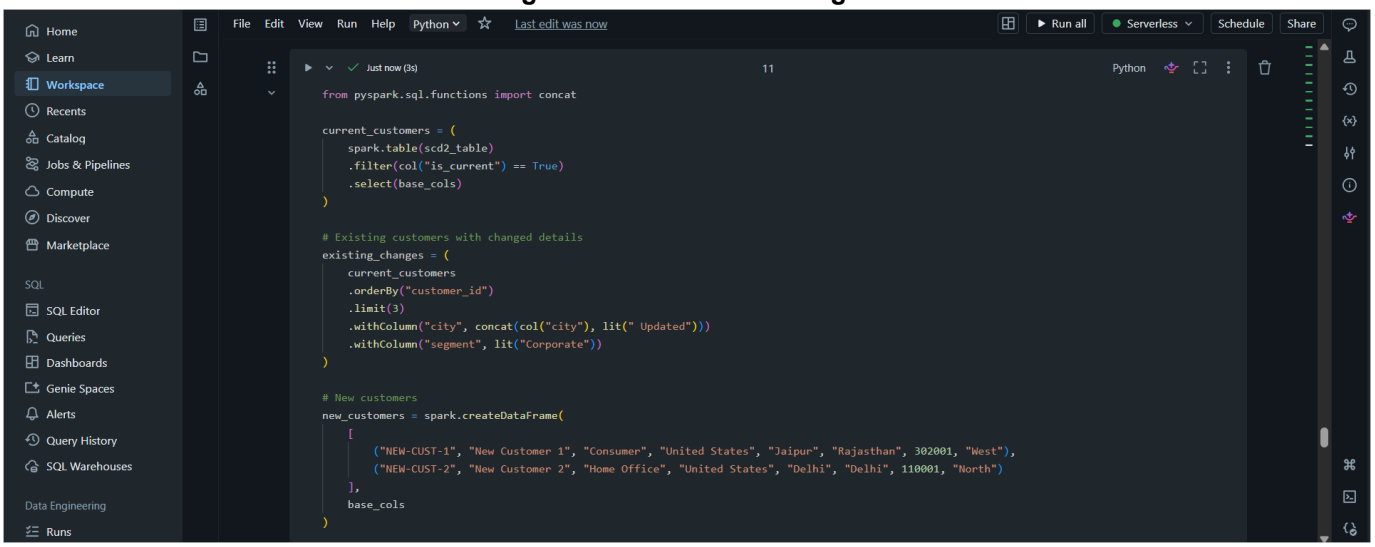


Figure 16: Incremental Customers

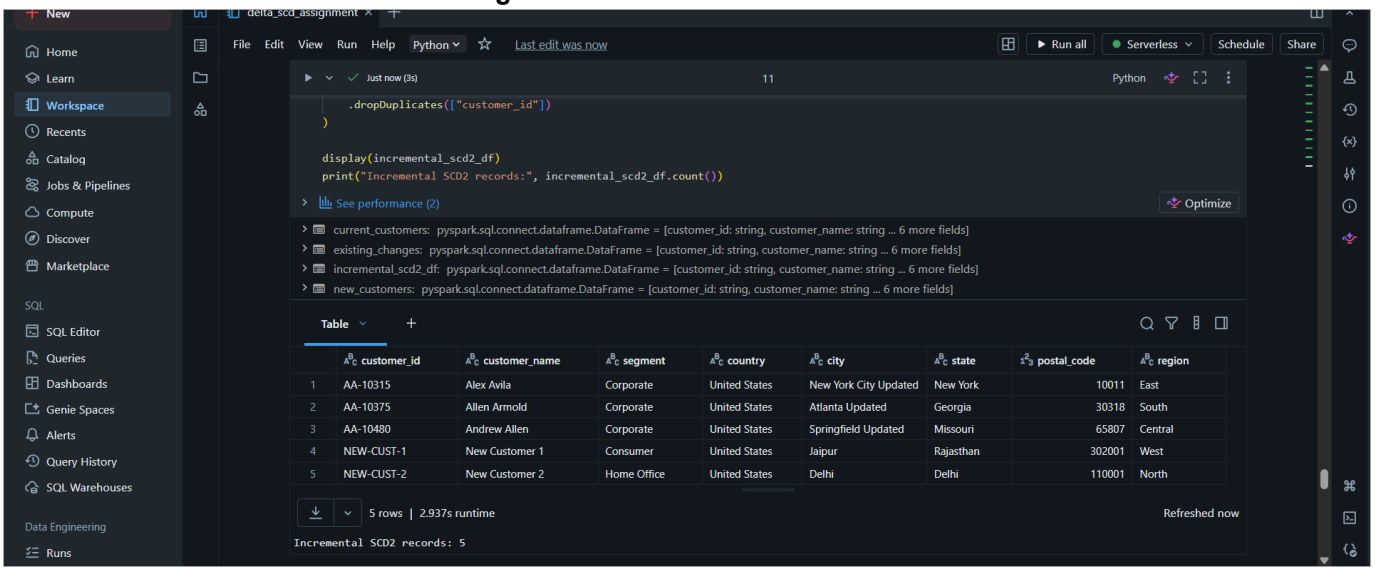


Figure 17: Hash Check

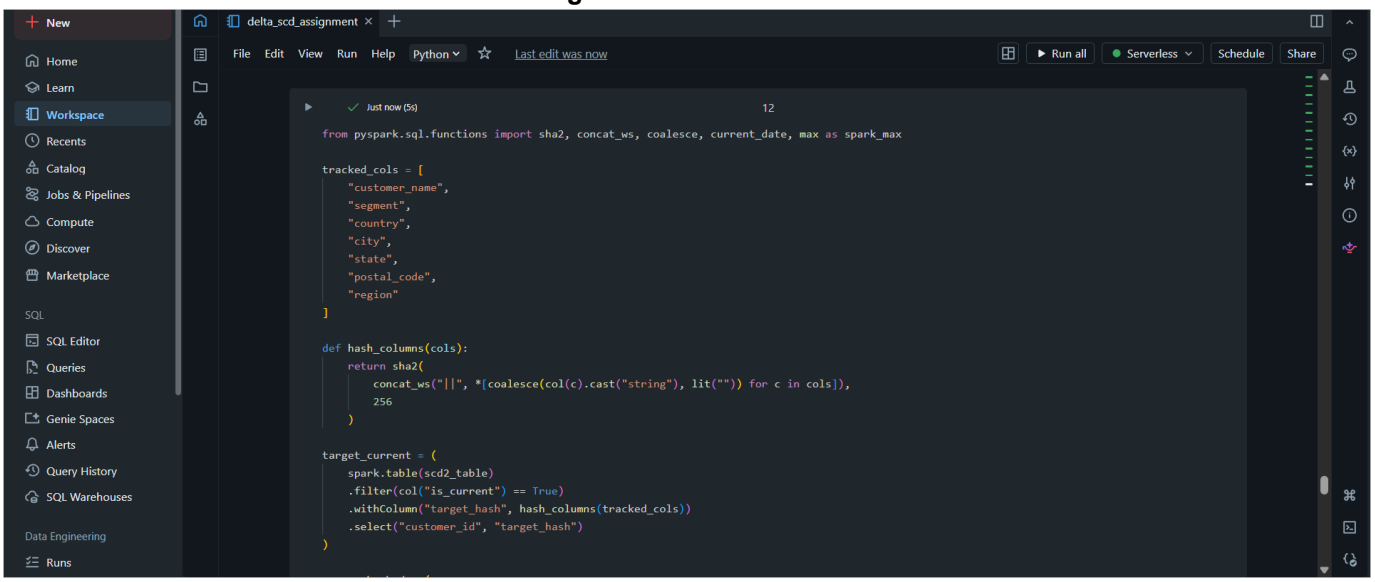


Figure 18: Insert Records

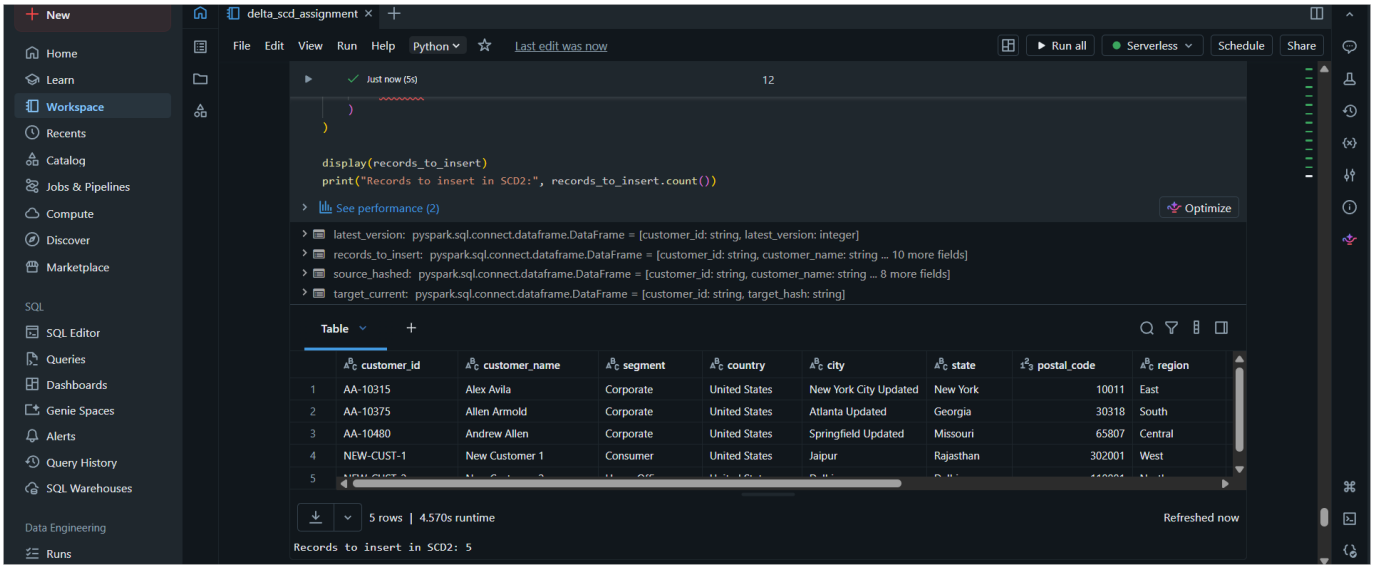


Figure 19: Expire Records

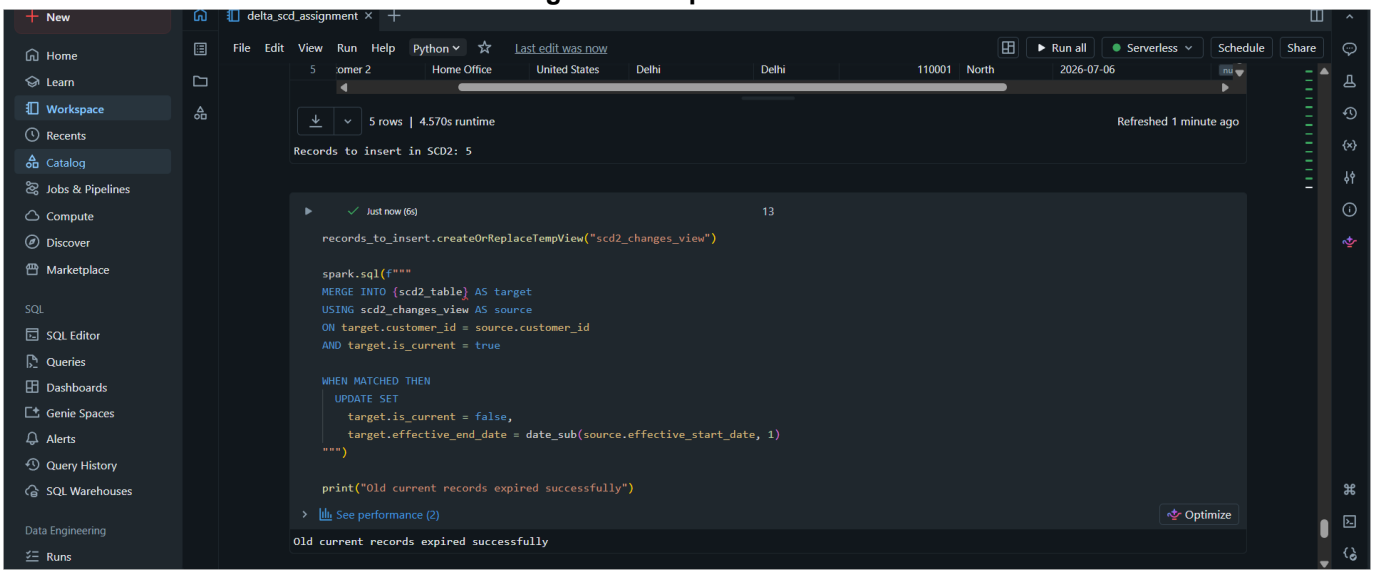


Figure 20: New Versions

Figure 21: Final SCD2

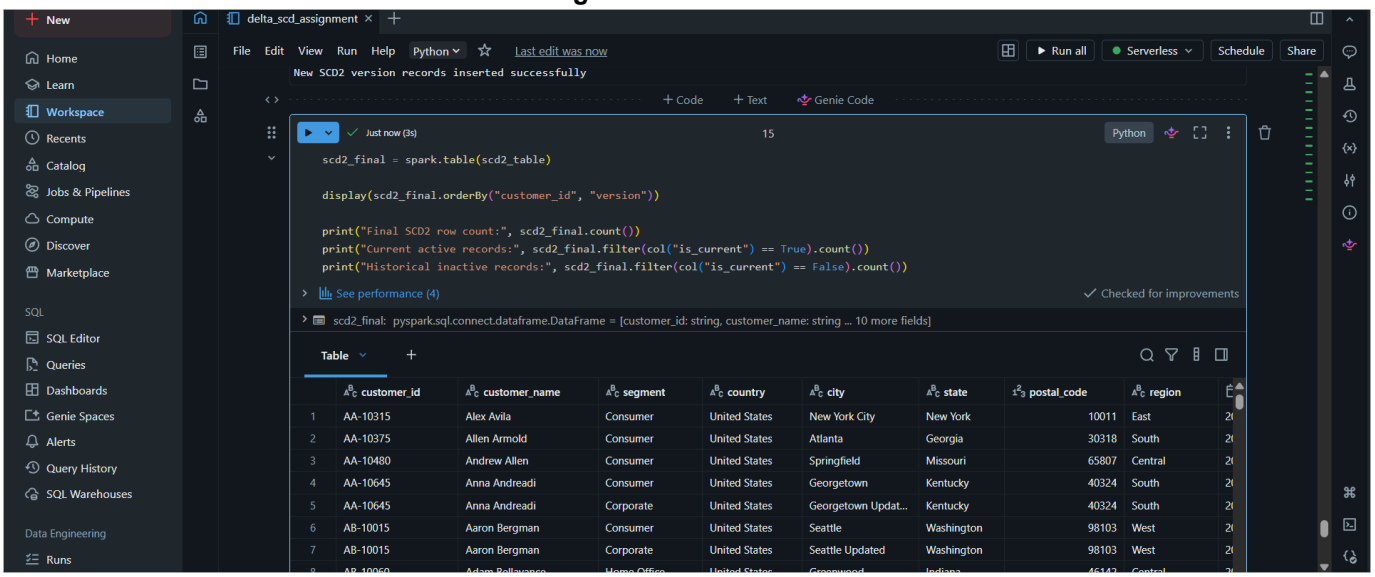


Figure 22: SCD2 Count

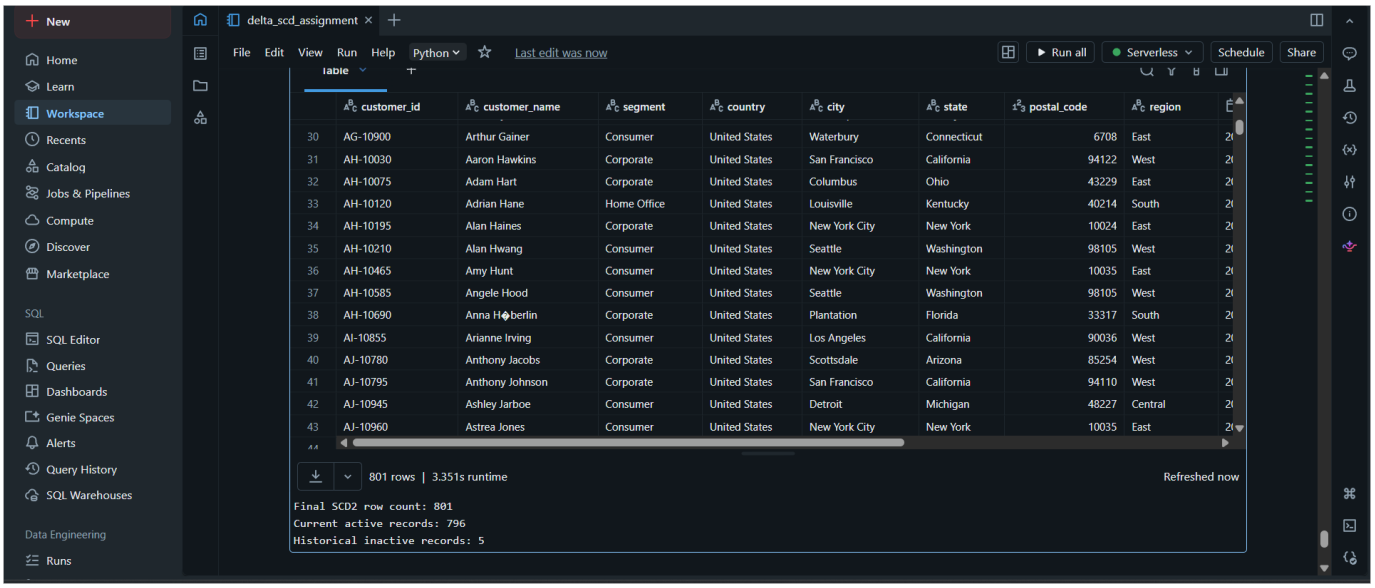


Figure 23: Current Validation

Figure 24: Duplicate Result

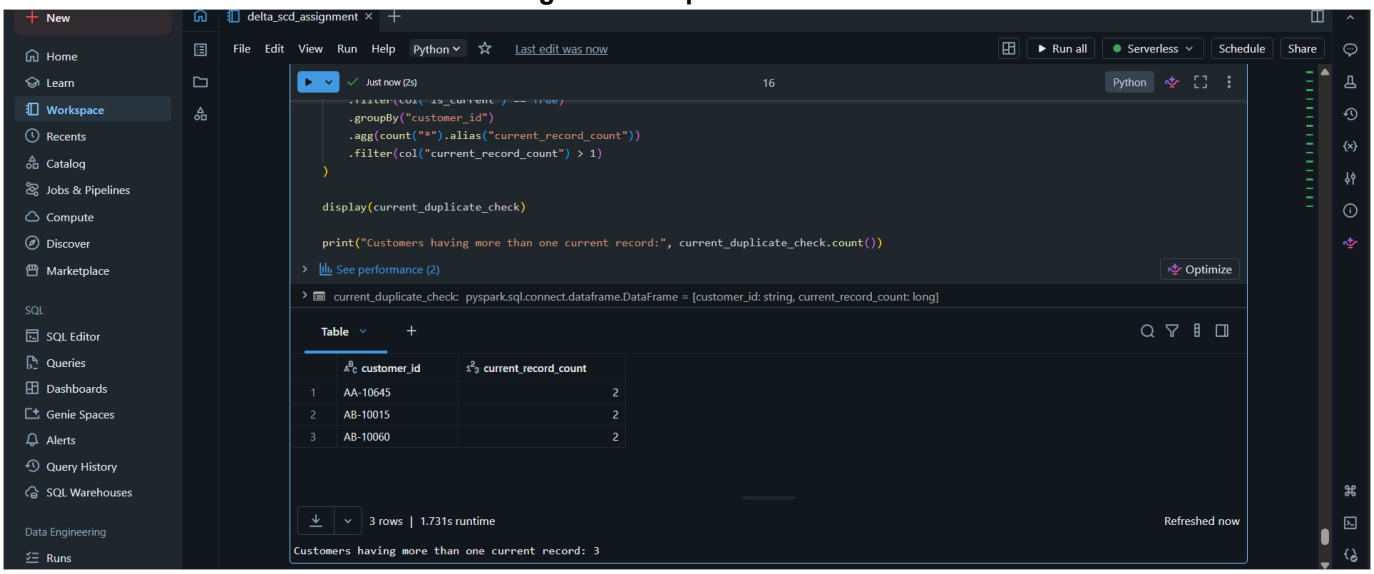


Figure 25: History Records

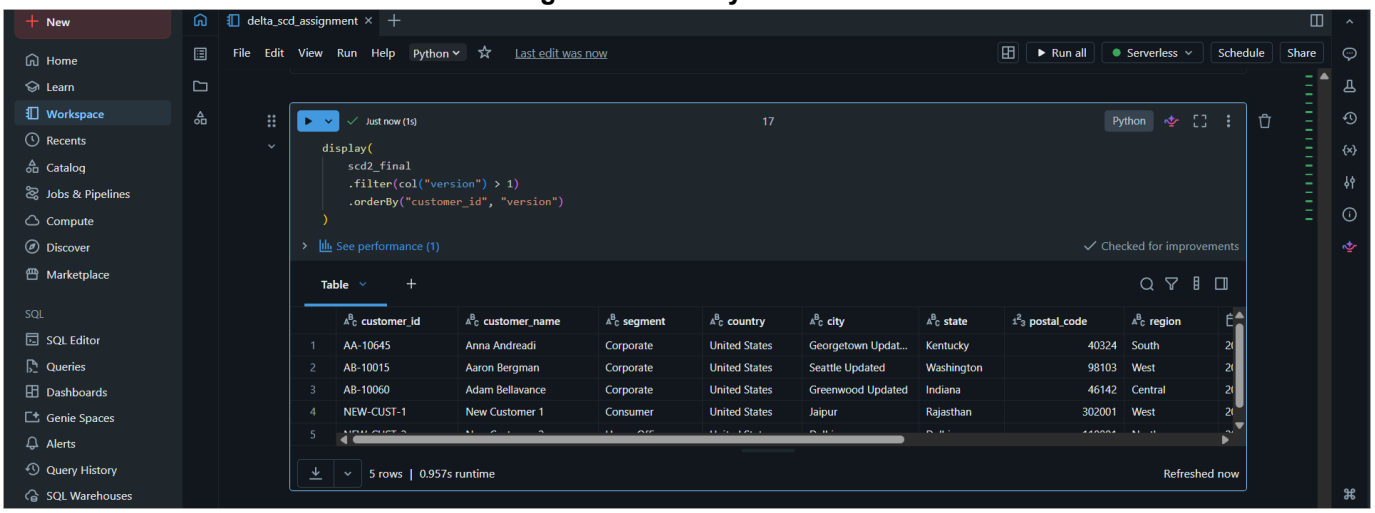


Figure 26: Final Summary

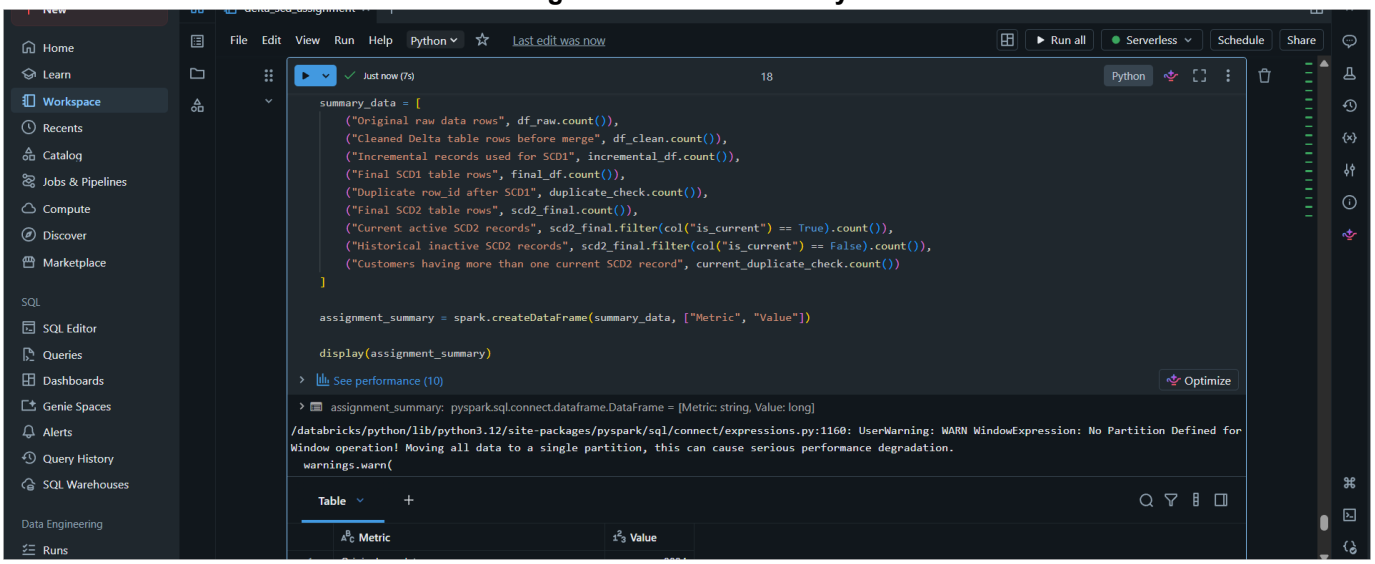
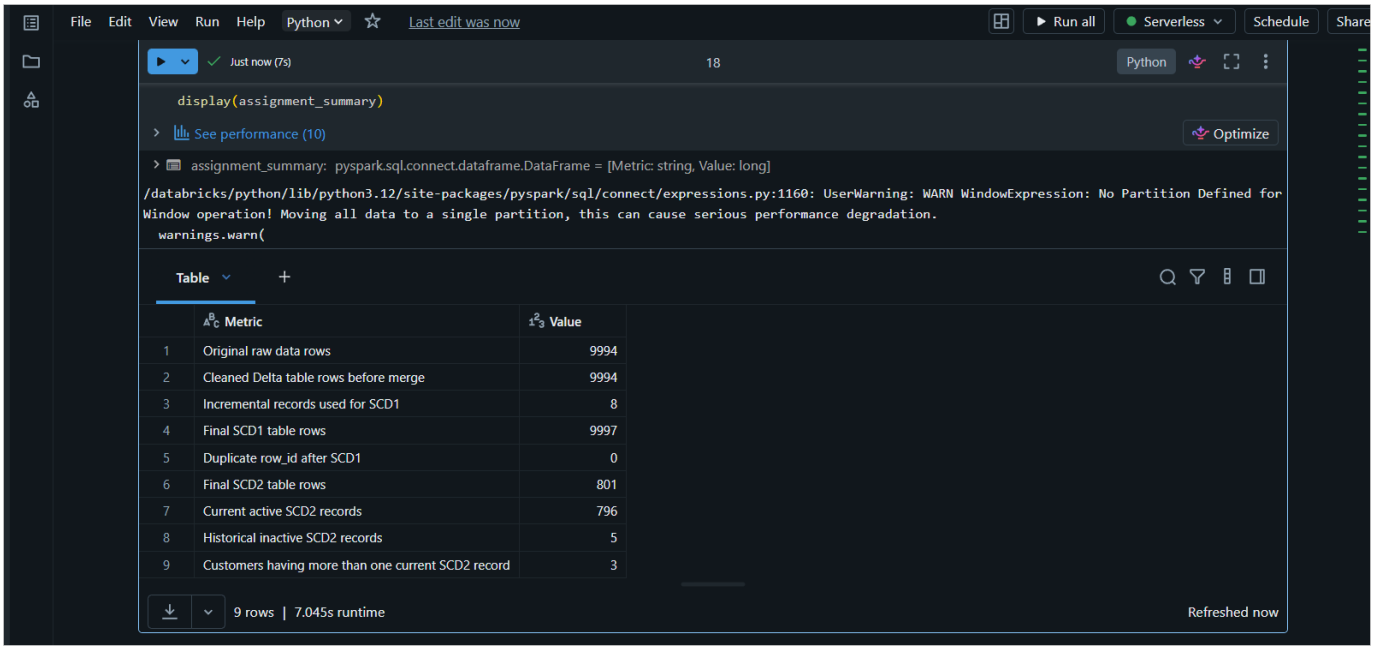


Figure 27: Summary Output



Short explanation

In this assignment, I loaded the Superstore dataset in Databricks and cleaned it by handling null values, removing duplicates, and fixing column names. After cleaning, saved the data as a Delta table. Then I created incremental records and used the Delta Lake MERGE command to update existing rows and insert new rows. Also validated the final output using row count and duplicate checks. SCD Type 2 was also added to maintain historical customer records.